

4.2 HW

- 4.2.1 a) $(1,1) \notin L$ because $1 \neq 1$: not reflexive
 $x < x$ for any x value : anti-reflexive *
 $1 < 2$ but $2 \not< 1 \rightarrow$: not symmetric
 if $x < y$ and $y < x$ then $x = y$: anti-symmetric *
 if $x < y$ and $y < z$, then $x < z$: transitive *
- b) $(x, x) \in E$ because $x \leq x \forall x \in \mathbb{R}$: reflexive *
 $1 \leq 2$ but $2 \not\leq 1 \rightarrow$: not symmetric
 if $x \leq y$ and $y \leq x$ then $x = y$: anti-symmetric *
 if $x \leq y$ and $y \leq z$ then $x \leq z$: transitive *
- c) $x' = x \forall x$ making it : reflexive *
 $2^2 = 4$ but $4^{1/2} = 2$ and $\frac{1}{2} \notin \mathbb{Z}^+$: not symmetric *
 If $x^n = y$ and $y^n = x$, then $(y^n)^n = y$
 $n^2 = y$
 so if xPy and yPx then $x = y$
 because both can not be true. : Anti-symmetric *
 If xPy and yPx , then $x^n = y$ and $y^n = x$
 $x^{n^2} = y$ and $y^{n^2} = x$
 $x^{n^2} = z$ and $n^2 \in \mathbb{Z} \rightarrow nPx$: transitive *

4.3 HW + 4.5 HW

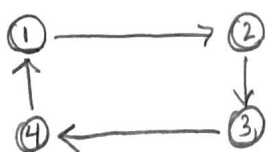
- 4.3.2
- yes it is a circuit, no it is not a cycle.
 - The longest cycle in the graph is $d \rightarrow b \rightarrow c \rightarrow g \rightarrow f \rightarrow d$ with a length 5
 - Cycle of length 4 is $g \rightarrow f \rightarrow e \rightarrow c \rightarrow g$
 - Example of length 5: $\langle a, b, c, g, f \rangle$
 - There are not self-loops in the graph.
 - $\langle a, g, f, c, d \rangle$ is not a walk, trail, or path within the graph because there is no edge from $f \rightarrow c$.
 - Yes it is a walk as well as a trail and path.
 - It is neither a circuit or a cycle.

4.5.2 Vertex $(V) = \{1, 2, 3, 4\}$ Edges $(E) = \{(1, 2), (1, 4), (4, 2), (3, 4), (2, 1)\}$

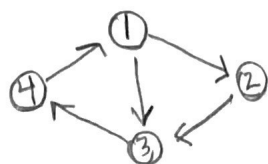
$G^2 =$



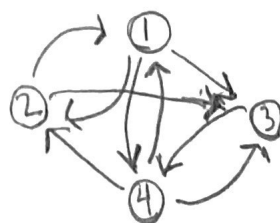
$G^3 =$



$G^4 =$



$G \cup G^2 \cup G^3 \cup G^4 \rightarrow G^+ =$



4.6 HW

4.6.1 a) The adjacency matrix is

$$G = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} \end{matrix}$$

$$G^2 = G * G$$

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$G^3 = G^2 * G$$

$$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$G^4 = G^3 * G$$

$$\begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

$$G^+ = G \cup G^2 \cup G^3 \cup G^4$$

$$= G + G^2 + G^3 + G^4$$

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} = G^+ \begin{bmatrix} 0 & 2 & 1 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 1 & 1 & 2 \\ 0 & 2 & 1 & 2 \end{bmatrix}$$

4.8 HW

4.8.2 $S = \{7, 2, 13, 44, 56, 34, 99, 31, 4, 17\}$

$X \in S/4$	Remainder
$7 = 7/4$	3
$2 = 2/4$	2
$13 = 13/4$	1
$44 = 44/4$	0
$56 = 56/4$	0
$34 = 34/4$	2
$99 = 99/4$	3
$31 = 31/4$	3
$4 = 4/4$	0
$17 = 17/4$	1

$$e_1(0) = \{44, 56, 4\}$$

$$e_1(1) = \{13, 17\}$$

$$e_1(2) = \{2, 34\}$$

$$e_1(3) = \{7, 99, 31\}$$

$$S = e_1(0) \cup e_1(1) \cup e_1(2) \cup e_1(3)$$

$e_1(0), e_1(1), e_1(2), e_1(3)$ are
they equivalence classes of $\mathcal{D}$.

4.9 HW (continued)

4.9.2 a) $a_{n+1} - a_n = (n+1)^2 - 2(n+1) - n^2 + 2n$
 $= n^2 + 1 + 2n - 2n - 2 - n^2 + 2n$
 $= 2n - 1$

$$n \geq 1$$

Increasing and non-decreasing

b) $a_{n+1} - a_n = (n+1)^2 - 3(n+1) - n^2 + 3n$
 $= n^2 + 1 + 2n - 3n - 3 - n^2 + 3n$
 $= 2n - 2$

$$2(1) - 2 = 0$$

$$2(2) - 2 = 2$$

increasing and non-decreasing

c) $a_{n+1} - a_n = (n+1)^2 - 4(n+1) - n^2 + 4n$
 $= n^2 + 1 + 2n - 4n - 4 - n^2 + 4n$
 $= 2n - 3$

$$n=1 \rightarrow 2(1) - 3 = -1$$

$$n=2 \rightarrow 2(2) - 3 = 1$$

Increasing & non-decreasing

d) $a_{n+1} - a_n = 2^{n+1} - (n+1)! - 2^n + n!$
 $= 2^n(2-1) - n!(n+1-1)$
 $= 2^n - n(n!)$

$$n=1 \rightarrow 2 - 1 = 1$$

$$n=2 \rightarrow 4 - 4 = 0$$

$$n=3 \rightarrow 8 - 18 = -10$$

non-increasing and decreasing

e) $a_{n+1} - a_n = 2^{n+1} - 3^{n+1} - 2^n + 3^n$
 $= 2^n(2-1) - 3^n(3-1)$
 $= 2^n - 3^n(2)$
 $= 2^n - 2 \cdot 3^n$

$$n=1 \rightarrow 2 - 6 = -4$$

$$n=2 \rightarrow 4 - 2 \cdot 9 = -16$$

decreasing and non-increasing

4.10 HW

4.10.1 a) $a_1 = 1$

$$a_2 = 2$$

$$a_3 = a_2 \cdot a_1 = 2 \times 1 = 2$$

$$a_4 = a_3 \cdot a_2 = 2 \times 2 = 4$$

$$a_5 = a_4 \cdot a_3 = 4 \times 2 = 8$$

$$a_6 = a_5 \cdot a_4 = 8 \times 4 = 32$$

The first six terms are 1, 2, 2, 4, 8, 32

b) $a_1 = 1, a_2 = 5$

$$a_3 = 2a_2 + 3a_1 = 2(5) + 3(1) = 13$$

$$a_4 = 2a_3 + 3a_2 = 2(13) + 3(5) = 36$$

$$a_5 = 2a_4 + 3a_3 = 2(36) + 3(13) = 111$$

$$a_6 = 2a_5 + 3a_4 = 2(111) + 3(36) = 330$$

First 6 terms are 1, 5, 13, 36, 111, 330

c) $g_1 = 2, g_2 = 1$

$$g_3 = 3 \cdot g_2 + g_1 = 3(1) + 2 = 5$$

$$g_4 = 4 \cdot g_3 + g_2 = 4(5) + 1 = 21$$

$$g_5 = 5 \cdot g_4 + g_3 = 5(21) + 5 = 110$$

$$g_6 = 6 \cdot g_5 + g_4 = 6(110) + 21 = 571$$

first 6 terms of g are 2, 1, 5, 21, 110, 571

4.10.1. d) $c_1 = 4$
 $c_2 = 5$

$$c_3 = c_2 \cdot c_1 = 5 \cdot 4 = 20$$

$$c_4 = c_3 \cdot c_2 = 20 \cdot 5 = 100$$

$$c_5 = c_4 \cdot c_3 = 100 \cdot 20 = 2000$$

$$c_6 = c_5 \cdot c_4 = 2000 \cdot 100 = 200,000$$

The first 6 terms are 4, 5, 20, 100, 2000, 200000

e) $b_1 = 1$
 $b_2 = 3$

$$b_3 = b_2 - 7(b_1) = 3 - 7(1) = -4$$

$$b_4 = b_3 - 7(b_2) = -4 - 7(3) = -25$$

$$b_5 = b_4 - 7(b_3) = -25 - 7(-4) = 4$$

$$b_6 = b_5 - 7(b_4) = 4 - 7(-25) = 179$$

The first 6 terms are 1, 3, -4, -25, 4, 179

4.11 HW

4.11.1 a) $\sum_k^4 = -1^k$

$$(-1)^2 + 0^2 + 1^2 + 2^2 + 3^2 + 4^2 \\ 1 + 1 + 4 + 9 + 16 = \boxed{31}$$

b) $\sum_0^4 = 0^{2^k} = 2^0 + 2^1 + 2^2 + 2^3 + 2^4 \\ = 1 + 2 + 4 + 7 + 16 = \boxed{31}$

c) $\sum_k^2 = -3k^3 = (-3)^3 + (-2)^3 + (-1)^3 + (0)^3 + (1)^3 + (2)^3 \\ = -27 - 8 - 1 + 0 + 1 + 8 \\ = \boxed{-27}$

d) $\sum_{k=0}^3 3^k \\ = 3^0 + 3^1 + 3^2 + 3^3 \\ = 1 + 3 + 9 + 27 = \boxed{40}$

e) $\sum_{k=0}^{200} = 2 + 400 + 60,300$ last term = 602
 $= \boxed{60,702}$ number terms = 201
 $\frac{201}{2} [2 + 602]$

f) $\sum_{k=0}^{200} 2 \times (1.01)^k \\ = 2 \sum_{k=0}^{200} (1.01)^k \\ = 2(1 + 1.01 + (1.01)^2 + \dots + (1.01)^{200}) \\ = 2 \left(\frac{1(1.01)^{201} - 1}{1.01 - 1} \right) \\ = 2 \left(\frac{7.38917803 - 1}{.01} \right) \\ = 200(6.38917803) \\ = \boxed{1277.83561}$

4.12 Hwl

4.12.1 a) $n=3$, then

$$\sum_{j=1}^3 j^2 = 1^2 + 2^2 + 3^2 = 14$$
$$= \frac{3(3+1)(6+1)}{6} = \frac{3(4)(7)}{6} = 14$$

b) $P(3)$ is true

$$P(k) = \frac{k(k+1)(2k+1)}{6}$$

$$c) P(k+1) = \frac{(k+1)(k+2)(2k+3)}{6}$$

d) for the base case, it must be proven that $P(1)$ is true.

e) k is a positive integer. To prove $P(n)$ will be true if proven that $P(k+1)$ is true.

f) The inductive hypothesis is that $P(k)$ is true for a positive integer k .